HIGHER DIRECT IMAGES ALONG A LIMIT OF MORPHISMS, FOR A COEFFICIENT COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the reduction to cohomology sheaves and the reassembly are both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The set-up of the companion result: inverse system of morphisms with affine transitions and the limit morphism. Same chapter, the statement of lemma-relative-colimit-general delivered verbatim as a context span in this card (source lines 7566-7582). Supplies the notation and the hypotheses, and the sheaf-level statement which the complex-level statement generalises.
- Spectral sequence or resolution argument comparing the cohomology sheaves of the pushforward. Same chapter, the technique by which a statement about sheaves is upgraded to bounded below complexes, together with the smoothness of the pullback functors. Supplies the reduction from the complex case to the sheaf case.

Source reference. Original lemma 01YV: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7573–7573. Navigation pointer invoked at proof line 7573; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05TC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7896–7896. Navigation pointer invoked at proof line 7896; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let I be a directed set. Let $g_i : X_i \to S_i$ be an inverse system of morphisms of schemes over I. Assume g_i is quasi-compact and quasi-separated and for $i' \geq i$ the transition morphisms $f_{i'i} : X_{i'} \to X_i$ and $h_{i'i} : S_{i'} \to S_i$ are affine. Let $g : X \to S$ be the limit of the morphisms g_i, see Limits, Section 01YV. Denote $f_i : X \to X_i$ and $h_i : S \to S_i$ the projections. Let $(\mathcal{F}_i, \varphi_{i'i})$ be a system of sheaves on $(X_i, f_{i'i})$. Set $\mathcal{F} = \operatorname{colim} f_i^{-1}\mathcal{F}_i$. Then*

$$R^p g_* \mathcal{F} = \operatorname{colim}_{i \in I} h_i^{-1} R^p g_{i,*} \mathcal{F}_i$$

for all $p \geq 0$.

Lemma 0.2. *Let I, $g_i : X_i \to S_i$, $g : X \to S$, f_i, g_i, h_i be as in Lemma 0.1. Let $0 \in I$ and $K_0 \in D^+(X_{0,étale})$. For $i \geq 0$ denote K_i the pullback of K_0 to X_i. Denote K the pullback of K_0 to X. Then*

$$R^p g_* K = \operatorname{colim}_{i \geq 0} h_i^{-1} R^p g_{i,*} K_i$$

for all $p \in \mathbf{Z}$.

Proof. Fix an integer $p_0 \in \mathbf{Z}$. Let a be an integer such that $H^j(K_0) = 0$ for $j < a$. We will prove the formula holds for all $p \leq p_0$ by descending induction on a. If $a > p_0$, then we see that the left and right hand side of the formula are zero for $p \leq p_0$ by trivial vanishing, see Derived Categories, Lemma 05TC. Assume $a \leq p_0$. Consider the distinguished triangle

$$H^a(K_0)[-a] \to K_0 \to \tau_{\geq a+1} K_0$$

Pulling back this distinguished triangle to X_i and X gives compatible distinguished triangles for K_i and K. For $p \leq p_0$ we consider the commutative diagram

$$\begin{array}{ccc}
\operatorname{colim}_{i \geq 0} h_i^{-1} R^{p-1} g_{i,*}(\tau_{\geq a+1} K_i) & \xrightarrow[\alpha]{} & R^{p-1} g_*(\tau_{\geq a+1} K) \\
\downarrow & & \downarrow \\
\operatorname{colim}_{i \geq 0} h_i^{-1} R^p g_{i,*}(H^a(K_i)[-a]) & \xrightarrow[\beta]{} & R^p g_*(H^a(K)[-a]) \\
\downarrow & & \downarrow \\
\operatorname{colim}_{i \geq 0} h_i^{-1} R^p g_{i,*} K_i & \xrightarrow[\gamma]{} & R^p g_* K \\
\downarrow & & \downarrow \\
\operatorname{colim}_{i \geq 0} R^p g_{i,*} \tau_{\geq a+1} K_i & \xrightarrow[\delta]{} & R^p g_* \tau_{\geq a+1} K \\
\downarrow & & \downarrow \\
\operatorname{colim}_{i \geq 0} R^{p+1} g_{i,*}(H^a(K_i)[-a]) & \xrightarrow{\epsilon} & R^{p+1} g_*(H^a(K)[-a])
\end{array}$$

with exact columns. The arrows β and ϵ are isomorphisms by Lemma 0.1. The arrows α and δ are isomorphisms by induction hypothesis. Hence γ is an isomorphism as desired. □